Hyuk Che (Luke) Kwon

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Education

University of Pennsylvania | Philadelphia, PA — Aug. 2025 - May 2027
- Master of Science in Engineering, Computer Graphics and Game Technology — GPA: 3.95

Sogang University | Seoul, South Korea — Mar. 2018 - Aug. 2024
- Bachelor of Science in Engineering, Computer Science and Engineering — Major GPA: 3.92
- Bachelor of Arts, Global Korean Studies — Magna Cum Laude

Programming Skills

Languages: C, C++, GLSL, Python, F#

Tools/Frameworks: CUDA, OpenGL, Nvidia Nsight, RenderDoc, GLM, Maya, Blender, Houdini, Git, Visual Studio, Qt

Experience

Software R&D Intern | *VS Graphics Team, LG Electronics* — Jun. 2026 - Aug. 2026
- Optimized CARLA, an Unreal Engine driving simulator for training autonomous vehicles in virtual environments
- Multithreaded JPEG decoding and removed redundant mipmaps for texture streaming, raising FPS from 11 to 20
- Diagnosed a GPU stall via Unreal Insights and implemented asynchronous GPU readback, raising FPS from 20 to 30
- Developed a Python script for dynamic client-side FPS pacing and diagnosed a memory leak in no-rendering mode

Teaching Assistant | *University of Pennsylvania* — Jan. 2026 - Present
- Assisted in teaching Introduction to Computer Graphics (CIS 5600) and Computer Animation (CIS 5620)
- Led review sessions, graded assignments, and held weekly office hours to help students with course material and projects

Projects

CUDA Boids Simulation | C++, CUDA, OpenGL, Nsight Systems — Sep. 2026
- Developed a CUDA flocking simulation with brute-force, scattered-grid, and coherent-grid neighbor search algorithms
- Replaced all-pairs search with uniform-grid partitioning and contiguous boid storage for efficient neighbor queries
- Profiled block sizes and grid granularities, raising kernel occupancy from 33% to 83% and simulation throughput by 17%

Deferred Renderer with Screen Space Reflections | C++, OpenGL, GLSL, Nsight Graphics — Apr. 2026
- Built a deferred rendering pipeline storing surface attributes to a multi-attachment G-buffer, achieving a 4.4× speedup over forward rendering on scenes with many lights and heavy overdraw
- Implemented SSR via G-buffer ray marching, with multi-level Gaussian blur for varying surface roughness
- Precomputed diffuse and split-sum specular IBL convolutions, yielding a 5.1× speedup at higher sample quality

Monte Carlo Path Tracer | C++, OpenGL, GLSL — Feb. 2026
- Developed a Monte Carlo path tracer solving the Light Transport Equation for diffuse, specular, and dielectric materials
- Built a Multiple Importance Sampling integrator using the Power Heuristic to combine direct BSDF and light sampling
- Implemented a full global illumination integrator computing MIS direct lighting at every bounce with BSDF-sampled indirect lighting and environment map support

Mini-Minecraft | C++, OpenGL, GLSL, RenderDoc — Dec. 2025
- Implemented player physics by incorporating ray marching to support collision detection on spherical environments
- Sampled 3D Perlin noise for cave systems and built a post-processing pipeline for underwater effects
- Created a procedural skybox using 3D Worley noise to generate stars and 3D Perlin noise-based FBM for nebulae